

Enclosure 5

**Near Earth Orbit Network (NEON)
Sounder for Microwave-Based Applications
(SMBA)
Contractor Work Breakdown Structure
(CWBS)**

July 2025

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1. DESCRIPTION / PURPOSE

The CWBS establishes a product-oriented framework for reporting program cost, schedule, and technical performance and provides a basis for uniform planning, reporting status, program visibility, and assignment of responsibilities. Additionally, the CWBS provides a product-oriented logical subdivision of hardware, software, services, facilities, etc., that contribute to and/or make up the total project scope of effort and work. The CWBS dictionary provides a narrative description of the work tasks and effort to be performed in each CWBS element.

2. PREPARATION INFORMATION

Contents:

The CWBS and CWBS dictionary are two distinct project documents used for defining the approved project scope of work. The contents of each document are detailed in the following paragraphs:

CWBS – A logical, hierarchical display of the subdivision of all project work to be completed.

The CWBS shall include the approved element title and element number.

CWBS dictionary – The CWBS dictionary shall briefly describe and document the work content of every CWBS element and relevant efforts associated with each element (e.g., design, development, and manufacturing).

The CWBS levels shall be governed by the Following. The actual letter and number designations can be adapted to meet the needs of the Contractor's systems

Level 1 - Total contract effort

Level 2 Definitions: Major contract activity as described below

- A. Engineering Unit / Non-Recurring Activities – All effort to plan and manage the design of the instrument including detailed design activities, design reviews, risk reduction (engineering models, etc), software development, ground support equipment (GSE) development, verification planning, etc.
- B. Flight Model (FM) -1 – All effort related to fabrication, assembly, test, launch operations and on-orbit testing of the satellite
- C. FM-2
- D. FM-3
- E. FM-4
- F. FM-5
- G. FM-6
- H. FM-7
- I. FM-8
- J. Reserved
- K. Reserved
- L. Mission Operations - Long term mission support including any management, business management support, systems engineering etc that is not unique to a particular mission

Level 3 Definitions (Note: X denotes the Level 2 Contract Activity) – the work content as described is not necessarily all inclusive.

- X.01 Program Management – Including program leadership, resource management, program planning (schedule), data management, configuration management, support for major design reviews
- X.02 Systems Engineering - Including system engineering leadership, interface management (instrument, launch vehicle, ground), system specification definition and flow-down, verification planning, system level analysis and modeling
- X.03 Mission Assurance - Including mission assurance leadership, quality engineering, subcontract surveillance, safety, reliability, parts and materials management
- X.04 Hardware / Software – All subsystem hardware and software effort including parts and materials which should be accounted for in a separate line. Effort included in each level 2 WBS
- X.05 Integration & Test - Including instrument baseline testing and environmental testing and post environment test activities including instrument packing and shipping.
- X.06 GSE – design, fabrication, procurement assembly, test and maintenance of electrical, mechanical GSE and mission simulation and GSE software
- X.07 Post Delivery Support (PDS) – Any support required after the Pre-Ship Review/Pre-Storage Review including satellite integration and testing and if needed storage costs or testing during storage prior to shipping to spacecraft vendor
- X.08 Launch and Post Launch Testing– support for pre-launch planning, launch site support, support for mission unique or special launch services, launch-vehicle integration (spacecraft portion), and the first 6 months of post-launch testing prior to the handoff to NOAA.
- X.09 Reserved
- X.10 Reserved
- X.11 Long Term Support – Long term mission support including any management, business management support, systems engineering etc. that is not unique to a particular mission

Level 4 – The Contractor shall define WBS level within each level 3 and provide a brief description of the work content.